

Wk 0.4 General pre-prep log

Lab log 15/01/2026

Added recommended textbook to bibliography (Christensen and Moloney 2005).

Lab log 19/01/2026

Briefly discussed lab w/Dr Turci; he noted the “percolation” chapter in (Christensen and Moloney 2005) as a good starting point since it introduces many ideas relating to phase transitions that will be useful for this project

Blackboard resources “ising1.pdf” have provided a general overview of background concepts and information. Useful ideas and concepts summarised in “0.5 background theory and concepts.” These are NOT the full background theory for this lab, but represent a useful starting point.

Lab log 21/01/2026

Called lab partner to introduce self; went over basic logistics and plans, are now on same page.

Bb materials “earmark” list:

- “ising1.pdf”, “ising2.pdf” - recaps basic stat mech theory, and covers basic setup of ising model
- “ising exercise.ppt” - notes metropolis algorithm, alongside some observables to compute
- “finite-size-effects.pdf” - goes over phase transition details - order parameters and critical exponents

Looking through more of the BB materials; and trying to think of vague plan:

- Go through BB “finite size effects.pdf”
- Go through “percolation” chapter (Christensen and Moloney 2005) - (in retrospect this may take a while) - note section 1.8 supposedly outlines extraction of critical exponents
- Familiarize self w/python simulation provided
- Refactor code in a .py file - jupyter notebooks are more annoying to work with in my experience
 - Additional benefit: lets me use my personal library ([NayanGeneralUtils](#)) which may be useful
- Refactor/expand code “architecture” generally - some potential ideas:

- MAYBE rework simulation to allow arbitrary dimensionality (I.E, 1D, 2D, 3D, 4D...)
 - suspect there may be interesting comparisons for different dimensionalities
- Rework as object oriented - each config could be it's own object with observable methods (I.E, computeEnergy, magnetisation...) and simulation step method - a "simulation run" class could then act as a wrapper for many config objects over time
 - * May also need to handle different simulations "in parallel" - I.E, need to vary sim parameter to generate data points
 - * Maybe set it up s.t configuration is object w/update method; overwrites itself; is a data destructive idea, however should only need a final "equilibrium" configuration for a given set of sim parameters
 - Then a "dataset" object would be a list of final configs + some metadata (I.E, lattice size, # sim steps, dimensionality...)
- Export/import method - given the computational nature of this lab, heavy compute runs may occur - being able to save raw simulation runs data in a standard format may save compute time
- Additional observable computation - current code computes energy and magnetisation
 - adding additional observables may be useful for investigation, I.E:
 - * Heat capacity
 - * Susceptibility
 - * Helmholtz free energy
- Maybe setup shared github repo w/lab partner

Christensen, Kim, and Nicholas R Moloney. 2005. *Complexity and Criticality*. Imperial College Press. <https://doi.org/10.1142/p365>.